

I. Comprehensive Ledger of Major Cross-Chain Bridge Attacks (2020–2026)

This ledger details the most catastrophic structural and cryptographic failures in cross-chain bridge architecture, now updated to include the recent Taiko exploit.

- **Poly Network (August 2021) — \$612 Million:** An attacker exploited an access control vulnerability in the smart contract's `verifyHeaderAndExecuteTx` function, allowing them to overwrite the keeper role by brute-forcing a specific method string.
 - *Reference:* [<https://hackenproof.com/blog/web3-bridge-hacks>] (<https://hackenproof.com/blog/web3-bridge-hacks>)
- **Wormhole (February 2022) — \$326 Million:** An attacker bypassed signature verification on Solana by spoofing the `sysvar` account, allowing forged cross-chain validation that minted unbacked wrapped ETH.
 - *Reference:* [<https://www.helius.dev/blog/solana-hacks>] (<https://www.helius.dev/blog/solana-hacks>)
- **Ronin Network / Axie Infinity (March 2022) — \$624 Million:** Attackers compromised the private keys of five out of nine validator nodes through targeted social engineering, exceeding the multi-sig threshold required to authorize fraudulent withdrawals.
 - *Reference:* [<https://hackenproof.com/blog/web3-bridge-hacks>] (<https://hackenproof.com/blog/web3-bridge-hacks>)
- **Harmony Horizon Bridge (June 2022) — \$100 Million:** Attackers successfully compromised the private keys of the multi-signature wallet securing the bridge protocol.
 - *Reference:* [<https://koinly.io/blog/biggest-defi-hacks/>] (<https://koinly.io/blog/biggest-defi-hacks/>)
- **Nomad Bridge (August 2022) — \$190 Million:** A critical initialization error (`confirmAt[0] = 1`) during a smart contract upgrade meant the zero-hash (0x00) was accepted as a valid Merkle root. This allowed any user to spoof processing proofs and withdraw funds.
 - *Reference:* [<https://arxiv.org/html/2501.03423v1>] (<https://arxiv.org/html/2501.03423v1>)
- **BNB BSC Token Hub (October 2022) — \$586 Million:** A vulnerability in the IAVL tree proof verification allowed attackers to forge arbitrary messages and mint BNB directly on the BNB Smart Chain.
 - *Reference:* [<https://koinly.io/blog/biggest-defi-hacks/>] (<https://koinly.io/blog/biggest-defi-hacks/>)

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- **Multichain (July 2023) — \$231 Million:** A systemic failure in the centralized Multi-Party Computation (MPC) architecture led to compromised keys and massive unauthorized outflows across multiple connected networks.
 - *Reference:* https://koinly.io/blog/biggest-defi-hacks/
- **Orbit Bridge (December 2023) — \$81 Million:** An unauthorized access exploit targeting the protocol's multi-signature infrastructure, resulting in drained liquidity right before the close of the year.
 - *Reference:* https://arxiv.org/html/2410.14493v2
- **Cetus (2025) — \$260 Million:** A systemic exploit targeting smart contract logic and liquidity pool mathematical invariants on the Sui network, allowing attackers to artificially manipulate pricing and extract value.
 - *Reference:* https://koinly.io/blog/biggest-defi-hacks/
- **KelpDAO (2025/2026) — \$292 Million:** An attacker compromised a third-party Decentralized Verifier Network (DVN) linked to the protocol's LayerZero configuration, spoofing collateral deposits on one chain to unlock real funds on another.
 - *Reference:* https://www.visualcapitalist.com/sp/ig02-biggest-crypto-hacks-since-2025/
- **Hyperbridge (April 2026) — \$2.5 Million:** A validation gap allowed an attacker to utilize forged cross-chain messages to gain control of a wrapped DOT token contract, allowing arbitrary minting.
 - *Reference:* https://linch.com/blog/post/the-biggest-bridge-hacks-in-2026
- **Verus-Ethereum Bridge (May 2026) — \$15+ Million:** An attacker bypassed the protocol due to a missing source-amount validation in the underlying Solidity logic, successfully deceiving the bridge into executing cross-chain instructions without proper ECDSA validation.
 - *Reference:* https://linch.com/blog/post/the-biggest-bridge-hacks-in-2026

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- - **Taiko Bridge Exploit (June 2026) — \$1.7 Million:** A systemic trust-model failure where an RSA-3072 private key (`enclave-key.pem`) was publicly exposed in a GitHub repository (`taikoxyz/raiko`). The attacker utilized this key—which acted as the root of trust for the Intel SGX prover path via the `MRSIGNER` check—to register a rogue prover through `SgxVerifier.registerInstance`. The rogue hardware generated forged attestations for fake L2 state, deceiving the bridge into marking fraudulent withdrawals as `RETRIABLE` via `processMessage()` and releasing funds via `retryMessage()`.
 - - *References:* [<https://caution.co/blog/taiko-bridge-exploit.html>] (<https://caution.co/blog/taiko-bridge-exploit.html>) | [<https://lnkd.in/gJPqHEyT>] (<https://lnkd.in/gJPqHEyT>)